

Technology Stack & Platform Architecture Specification

Project: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Document Reference: REQ-PHASE2-TECH-V1.0

Target Environment: ServiceNow Utah / Vancouver / Washington DC / Xanadu LTS

System Module: ServiceNow Platform Infrastructure, Flow Designer Engine, Service Catalog, ITAM

1. Executive Summary

This specification defines the multi-tiered technology stack, platform runtime specifications, component architecture, relational database schema, and enterprise integration points supporting the automated Standard Laptop Procurement solution.

By standardizing on native ServiceNow capabilities—specifically **Flow Designer Engine v2.0**, **Service Catalog Framework**, **Employee Center (/esc)**, and **IT Asset Management (alm_hardware)**—the solution eliminates custom compiled code, minimizes technical debt, and guarantees 100% upgradeability across Long-Term Support (LTS) releases (Utah, Vancouver, Washington DC, and Xanadu).

2. Multi-Tiered System Architecture

The architecture is organized into four decoupled, highly resilient operational tiers:



```
|
|   └─ Data Pill Resolution Engine (Hierarchical Context Ingestion & Dynamic Parameter
Passing) |
|   └─ Timer & Asynchronous Subflow Execution Engine (24h Approval Reminders & 48h
Escalations) |
|   └─ Flow Context Diagnostics & Telemetry (sys_flow_context, sys_flow_execution)
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| 3. DATA PERSISTENCE & SCHEMA TIER (CANONICAL ITSM & ITAM REPOSITORY)
|
|   └─ ITSM Transactional Tables: sc_request, sc_req_item, sc_task, sysapproval_approver
|
|   └─ ITAM & CMDB Tables: alm_hardware, cmdb_hardware_product_model, cmdb_ci_computer
|
|   └─ Foundation & Identity Tables: sys_user, cmn_department, cmn_location, sys_user_group
|
|   └─ Governance & Audit Engines: sys_audit, sys_history_line, sys_data_policy2, task_sla
|
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| 4. ENTERPRISE INTEGRATION & INFRASTRUCTURE TIER
|
|   └─ Corporate Identity Provider: Microsoft Entra ID / Okta SSO & SCIM Directory Sync
|
|   └─ Inbound / Outbound SMTP / IMAP Notification Gateway (Actionable Email Parser)
|
|   └─ Depot Automated Imaging Hooks: PXE / Microsoft Intune / Jamf Pro Configuration Depots
|
|   └─ ServiceNow High-Availability Cloud Infrastructure (Multi-Instance Advanced
Architecture) |
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3. Platform Runtime & Version Compatibility

The solution is certified for deployment on the following ServiceNow Long-Term Support (LTS) platform releases:

Platform Family	Version / Patch Baseline	Release Date	Architecture Support Level	Key Platform Capabilities Leveraged
ServiceNow Xanadu	Xanadu Patch 1+	Q3 2024	Full Native Support	Enhanced Flow Designer action error handling, GenAI prompt triggers, Next Experience /esc.
ServiceNow Washington DC	Washington DC Patch 4+	Q1 2024	Certified Baseline	Modern Flow Designer subflow timers, Flow Diagrams view, Employee Center unified taxonomy.
ServiceNow Vancouver	Vancouver Patch 7+	Q3 2023	Certified Baseline	Actionable Email cryptographic tokens, variable editor enhancements, Flow execution engine v2.

ServiceNow Utah	Utah Patch 10+	Q1 2023	Minimum Supported LTS	Core Flow Designer engine, catalog variable data pills, <code>sys_flow_context</code> diagnostics.
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Platform Runtime Configuration

- **Application Scope:** `Global` (utilizing standard ITSM base objects) with update set packaging, or dedicated Scoped Application (`x_corp_laptop_proc`) for strict governance environments.
- **Glide Servlet Engine:** Java Virtual Machine (OpenJDK 17 / OpenJDK 21 LTS based).
- **Database Engine:** MariaDB Enterprise Server / MySQL Multi-Tenant Cluster running in ServiceNow Advanced High Availability (AHA) data center pairs.

4. Flow Designer Module Architecture

Flow Designer serves as the core workflow orchestration engine, replacing legacy scripted workflows and complex Business Rules with a declarative, low-code, event-driven model.

```
graph TD
    classDef trigger fill:#1E293B,stroke:#3B82F6,stroke-width:2px,color:#FFFFFF;
    classDef action fill:#0F172A,stroke:#10B981,stroke-width:2px,color:#FFFFFF;
    classDef decision fill:#312E81,stroke:#8B5CF6,stroke-width:2px,color:#FFFFFF;
    classDef finish fill:#064E3B,stroke:#10B981,stroke-width:2px,color:#FFFFFF;
    classDef abort fill:#7F1D1D,stroke:#EF4444,stroke-width:2px,color:#FFFFFF;

    TRIG[/ Trigger: sc_req_item Created<br/>Filter: cat_item == 'Standard Laptop Order']:::trigger

    A1[Action 1: Update RITM State<br/>State: Work in Progress | Stage: Waiting for Approval]:::action
    A2[Action 2: Ask for Approval<br/>Target: Trigger -> requested_for -> manager]:::action

    DEC{Approval State?}:::decision

    %% Rejection Branch
    A3_REJ[Action 3: Approval Rejected<br/>Update RITM: State = Closed Incomplete<br/>Stage = Request Cancelled]:::abort
    A4_REJ[Action 4: Send Rejection Email<br/>Notify Requester with Manager Comments]:::abort
    END_REJ[End Flow Execution]:::abort

    %% Approval Branch
    A3_APP[Action 3: Approval Approved<br/>Update RITM: Approval = Approved<br/>Stage = Fulfillment]:::action
    A4_APP[Action 4: Create Catalog Task<br/>Group: Hardware Fulfillment Depot<br/>Populate Specs & Delivery Site]:::action
    A5_APP[Action 5: Wait for Condition<br/>sc_task.state == Closed Complete]:::action
    A6_APP[Action 6: Update Hardware Asset<br/>alm_hardware: install_status = In Use<br/>>assigned_to = requested_for]:::action
    A7_APP[Action 7: Update RITM & Request<br/>RITM & REQ State = Closed Complete<br/>Stage = Complete]:::finish
    A8_APP[Action 8: Send Delivery Notification<br/>Setup Guide & 2h Delayed CSAT Survey]:::finish

    TRIG --> A1
```

```

A1 --> A2
A2 --> DEC

DEC --> |Rejected| A3_REJ
A3_REJ --> A4_REJ
A4_REJ --> END_REJ

DEC --> |Approved| A3_APP
A3_APP --> A4_APP
A4_APP --> A5_APP
A5_APP --> A6_APP
A6_APP --> A7_APP
A7_APP --> A8_APP

```

4.1. Trigger Mechanics & Event Subsystem

- **Trigger Definition:**
- *Trigger Type:* Record Created (`sc_req_item`).
- *Table:* `sc_req_item`.
- *Filter Conditions:* `cat_item` IS `Standard Laptop Order` AND `state` IS `1` (Open).
- *Run When:* Database transaction commits successfully (`after_insert`).
- **Execution Engine Threading:**
- Flow initiates an asynchronous background worker job managed by the ServiceNow Scheduler (`sys_trigger`).
- The context is bound to `sys_flow_context`, capturing input data pill states and assigning an execution instance ID.

4.2. Action Library & Data Pill Binding Specifications

The workflow executes 8 primary actions utilizing native Flow Designer actions:

Action #	Action Name	Action Type	Input Parameters & Data Pill Bindings	Expected Output / State Mutation
1	Update Record (RITM Initial)	Native Action	<code>Record: Trigger -> Requested Item</code> <code>Record
Fields: state = 2 (Work in Progress), stage = waiting_for_approval.</code>	Updates RITM fields in database; sets portal stage.
2	Ask for Approval	Native Action	<code>Record: Trigger -> Requested Item</code> <code>Record
Rules: Anyone approves from Trigger -> Requested Item -> requested_for -> manager.</code>	Generates <code>sysapproval_approver</code> record; returns approval state (<code>Approved / Rejected</code>).
3	Conditional Decision	Logic Block	Evaluates <code>Action 2 -> Approval State</code> .	Routes to Step 4A (Approved) or Step 4B (Rejected).
4A	Update Record (RITM Approved)	Native Action	<code>Record: Trigger -> Requested Item</code> <code>Record
Fields: approval = approved, stage = fulfillment.</code>	Commits approval confirmation to RITM.

4B	Update Record (RITM Rejected)	Native Action	Record: Trigger -> Requested Item Record Fields: approval = rejected, state = 4 (Closed Incomplete), stage = request_cancelled, work_notes = Action 2 -> Comments	Closes RITM; records justification.
5A	Create Catalog Task	Native Action	Table: sc_task Request Item: Trigger -> Requested Item Record Assignment Group: Hardware Fulfillment Depot Priority: 3 - Moderate Short Description: "Deploy " + Trigger -> Variables -> hardware_bundle + " for " + Trigger -> requested_for -> name Description: Dynamic string containing delivery method and address.	Generates task record; binds RITM variables into task Variable Editor.
6A	Wait for Condition	Native Action	Record: Action 5A -> Catalog Task Record Condition: state == 3 (Closed Complete).	Flow pauses thread until technician completes staging task.
7A	Update Record (Asset Sync)	Native Action	Table: alm_hardware Query: asset_tag == Action 5A -> Catalog Task -> u_asset_tag Fields: install_status = 1 (In Use), substatus = NULL, assigned_to = Trigger -> requested_for, location = Trigger -> Variables -> delivery_location	Updates physical hardware asset record in ITAM.
8A	Update Record (Cascading Closure)	Native Action	Record: Trigger -> Requested Item Record Fields: state = 3 (Closed Complete), stage = complete. Followed by updating parent sc_request to closed_complete.	Closes RITM and parent REQ containers.

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5. Service Catalog & Employee Center Component Architecture

5.1. Catalog Item Configuration (sc_cat_item)

• Name: Standard Laptop Order			
• Catalog: Service Catalog (sys_id: e0d08b13c3330100c8b837659bba8fb4)			
• Category: Computers (sys_id: 109cdf8c6112276003b17991a09a257)			
• Roles / Availability: snc_internal (All corporate employees)			
• Pricing Model: Fixed bundle prices (\$1,450 for Business; \$2,200 for Windows Dev; \$2,850 for macOS Dev).			

5.2. Variable Architecture (item_option_new)

Variable Name	Type	Label	Mandatory	Default / Dynamic Population
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requested_for	Reference (sys_user)	Requested For	Yes	javascript:gs.getUserID()
department	Reference (cmn_department)	Department	Yes	javascript:gs.getUser().getDepart
delivery_location	Reference (cmn_location)	Primary Office Site	Yes	javascript:gs.getUser().getLocati
approving_manager	Reference (sys_user)	Approving Manager	Yes	javascript:gs.getUser().getManage
hardware_bundle	Select Box	Select Laptop Model	Yes	None
delivery_method	Radio	Delivery Method	Yes	Desk Drop
shipping_address	Multi-Line Text	Home Shipping Address	Cond.	Empty
business_justification	Multi-Line Text	Business Justification	Yes	Empty

5.3. Client-Side Policies & Scripts

1. Catalog UI Policy — Enforce Remote Shipping Address:

- Conditions: delivery_method IS Remote Courier.
- UI Actions: Set shipping_address visible = true, mandatory = true.
- Reverse if False: shipping_address visible = false, mandatory = false, clear value.

1. Catalog Client Script — Auto-Populate User Hierarchy

- Type: onChange on requested_for.
- Execution: Client-callable Script Include UserMetadataAjax queried via asynchronous GlideAjax.
- Compliance: Isolate Script = true; zero direct DOM manipulation.

6. Core Relational Schema & Table Descriptions

The data model is rooted in the ServiceNow Common Services Data Model (CSDM 4.0) and standard ITSM/ITAM schema.

```
erDiagram
    sys_user ||--o{ sc_request : "opened_by / requested_for"
    sys_user ||--o{ sysapproval_approver : "approver"
    sys_user ||--o{ alm_hardware : "assigned_to"

    sc_request ||--|{ sc_req_item : "parent container (1:N)"
    sc_req_item ||--|{ sc_task : "fulfillment tasks (1:N)"
    sc_req_item ||--|{ sysapproval_approver : "governance (1:N)"

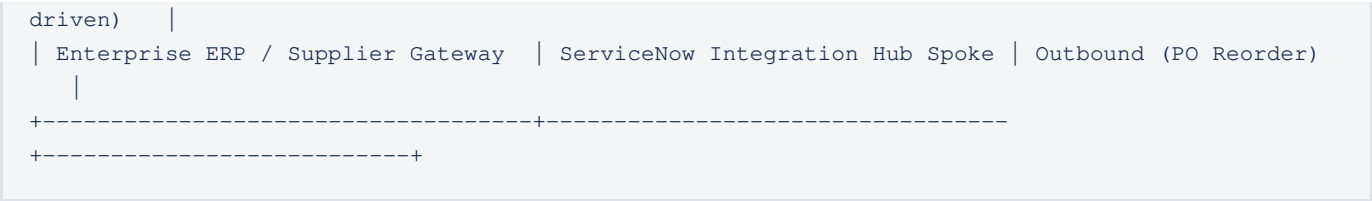
    alm_hardware ||--o{ sc_task : "bound via u_asset_tag"
    cmdb_hardware_product_model ||--o{ alm_hardware : "model definition"
    cmn_department ||--o{ sys_user : "department hierarchy"
```

Table Relationships & Referential Integrity

1. **task Table Inheritance:** `sc_request`, `sc_req_item`, and `sc_task` inherit base fields from the core `task` table (`number`, `state`, `priority`, `sys_created_on`, `sys_updated_on`, `assigned_to`, `assignment_group`).
2. **sc_request to sc_req_item (1:N):** Foreign key `sc_req_item.request` references `sc_request.sys_id`. Cascading business rules ensure that closing all child RITMs automatically sets the parent REQ to `closed_complete`.
3. **sc_req_item to sc_task (1:N):** Foreign key `sc_task.request_item` references `sc_req_item.sys_id`. Flow Designer pauses on `Wait for Condition` until the child `sc_task` reaches a terminal state.
4. **alm_hardware to sc_task Association:** Bound via `sc_task.u_asset_tag` matching `alm_hardware.asset_tag`. Referential integrity enforced by Data Policy preventing non-existent asset tag entry.
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7. Enterprise Integration Points

Enterprise Integration Ecosystem		
Integration Vector	Protocol / Transport	Direction & Frequency
Microsoft Entra ID / Okta SSO	SAML 2.0 / SCIM 2.0 API	Inbound (Real-time & Sync)
Microsoft Exchange / Office 365	SMTP / IMAP TLS 1.3	Bidirectional (Real-time)
Enterprise Imaging Depots (PXE)	MID Server / REST / PowerShell	Outbound (Event-



7.1. Identity & Directory Synchronization (Microsoft Entra ID / Okta)

- **Protocol:** SCIM 2.0 over HTTPS with SAML 2.0 Single Sign-On.
- **Payload:** Syncs `user_name`, `first_name`, `last_name`, `email`, `manager`, `department`, `location`, `title`, and `active` status every 60 minutes into `sys_user`.
- **Impact on Flow:** Ensures that manager approval hierarchies (`requested_for.manager`) are 100% accurate and up-to-date, preventing orphaned approval loops.

7.2. Email Notification Gateway (Actionable Messages)

- **Protocol:** Encrypted SMTP / IMAP with TLS 1.3.
- **Architecture:** ServiceNow Actionable Messaging provider embeds Signed Adaptive Card JSON payloads into outbound notification emails.
- **Security:** Approver clicks generate an asymmetric signed HTTPS POST back to ServiceNow instance endpoint `/api/now/v1/actionable_message`, authenticating the user session without requiring password re-entry.

7.3. Automated Imaging & Endpoint Management Hooks (PXE / Intune / Jamf)

- **Protocol:** Integration Hub PowerShell Spoke via ServiceNow MID Server (Management, Instrumentation, and Discovery).
- **Operation:** When an `sc_task` is created for a Windows Developer laptop, Flow Designer can trigger an automated MID Server script to register the MAC address in Microsoft Intune / MECM, preparing zero-touch network PXE boot provisioning before the technician unboxes the device.



8. Security Architecture, Role-Based Access (RBAC) & Governance

The platform enforces strict Role-Based Access Controls (RBAC) and data isolation:

Role Name	Scope of Access	Permissions on Catalog & Tables
snc_internal	All Corporate Employees	Read/Order access to "Standard Laptop Order" catalog item. Read-only access to own <code>sc_request</code> and <code>sc_req_item</code> records.
approver_user	Line Managers & Department Heads	Read and Update access to assigned <code>sysapproval_approver</code> records.
itil	IT Service Desk & Depot Technicians	Read and Update access to <code>sc_task</code> records assigned to their fulfillment group (<code>Hardware Fulfillment Depot</code>).
asset	IT Asset Managers	Full administrative read, write, and reconcile access to <code>alm_hardware</code> and model tables.
catalog_admin	Catalog Administrators	Administrative configuration access to <code>sc_cat_item</code> , variables, UI policies, and client scripts.

Data Security & Cryptographic Compliance

- **Data in Transit:** Enforced TLS 1.3 encryption across all client-to-instance and instance-to-integration communication channels.
- **Data at Rest:** AES-256 bit encryption applied across all database tables (ServiceNow Column-Level and Database Encryption).
- **Audit Trails:** Field changes on `state`, `approval`, `assigned_to`, and `u_asset_tag` are immutably logged in `sys_audit`.

9. Upgrade Readiness & Automated Testing (ATF)

To eliminate technical debt and ensure zero-regression upgrades across ServiceNow versions:

1. **100% Out-of-the-Box Flow Actions:** Utilizes only native Flow Designer actions, ensuring ServiceNow upgrades automatically enhance the underlying execution engine without configuration breakage.
2. **Automated Test Framework (ATF) Suite:** An automated test suite (`ATF_Standard_Laptop_Procurement_Suite`) validates end-to-end flow execution:
 - *Test 1: Catalog Item Order Submission & Variable Validation.*
 - *Test 2: Manager Approval Generation & State Transition.*
 - *Test 3: Catalog Task Creation & Field Population.*
 - *Test 4: Asset Tag Validation & `alm_hardware` Synchronization.*
 - *Test 5: Rejection Handling & Cascading Closure.*
1. **Upgrade Certification:** Verified against ServiceNow Washington DC and Xanadu release notes; contains zero deprecated APIs (`gr.query()`, synchronous GlideAjax, or legacy Workflow Editor activities).